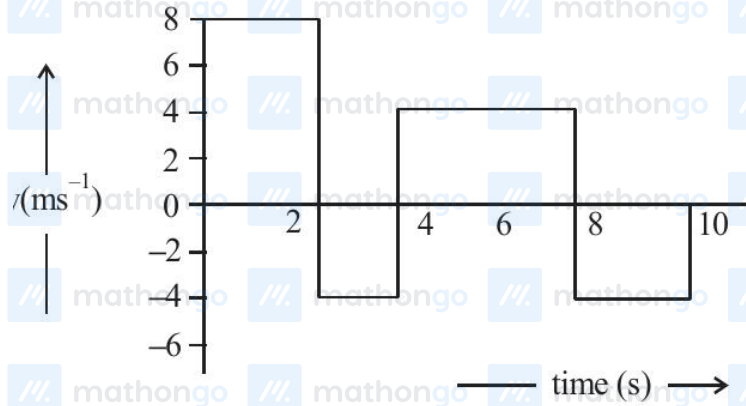


Q1 - 24 January - Shift 2

The velocity time graph of a body moving in a straight line is shown in figure.



Space for your notes:

The ratio of displacement to distance travelled by the body in time 0 to 10s is

- (1) 1:1
- (2) 1:4
- (3) 1:2
- (4) 1:3

Q2 - 25 January - Shift 1

A car travels a distance of 'x' with speed V_1 and then same distance 'x' with speed V_2 in the same direction. The average speed of the car is:

- (1) $\frac{v_1 v_2}{2(v_1 + v_2)}$
- (2) $\frac{v_1 + v_2}{2}$
- (3) $\frac{2x}{v_1 + v_2}$
- (4) $\frac{2v_1 v_2}{v_1 + v_2}$

Space for your notes:



Q3 - 25 January - Shift 2

The distance travelled by a particle is related to time t as $x = 4t^2$. The velocity of the particle at $t = 5$ s is .

- (1) 40 ms^{-1} (2) 25 ms^{-1}
(3) 20 ms^{-1} (4) 8 ms^{-1}

Space for your notes:

Q4 - 29 January - Shift 1

A tennis ball is dropped on to the floor from a height of 9.8 m. It rebounds to a height 5.0 m. Ball comes in contact with the floor for 0.2s. The average acceleration during contact is _____ ms^{-2} .

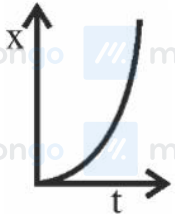
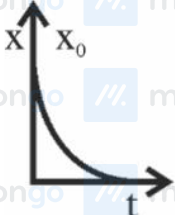
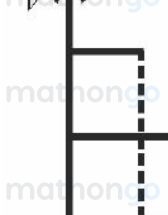
[Given $g = 10 \text{ ms}^{-2}$]

Space for your notes:

Q5 - 30 January - Shift 1

Match Column-I with Column-II :

Space for your notes:

Column-I (x-t graphs)		Column-II (v-t graphs)	
A.		I.	
B.		II.	
C.		III.	
D.		IV.	

Choose the correct answer from the options given below:

- (1) A- II B-IV, C-III, D-I
- (2) A- I. B-II, C-III, D-IV
- (3) A- II B-III, C-IV, D-I
- (4) A- I. B-III. C-IV, D-II

Q6 - 30 January - Shift 1

#MathBoleTohMathonGo

A horse rider covers half the distance with 5 m/s speed. The remaining part of the distance was travelled with speed 10 m/s for half the time and with speed 15 m/s for other half of the time. The mean speed of the rider averaged over the whole time of motion is $x/7$ m/s. The value of x is

Space for your notes:

Q7 - 30 January - Shift 2

An object is allowed to fall from a height R above the earth, where R is the radius of earth. Its velocity when it strikes the earth's surface, ignoring air resistance, will be :

Space for your notes:

- (1) $2\sqrt{gR}$ (2) $\sqrt{gR}$
(3) $\sqrt{\frac{gR}{2}}$ (4) $\sqrt{2gR}$

Q8 - 30 January - Shift 2

A vehicle travels 4 km with speed of 3 km/h and another 4 km with speed of 5 km/h, then its average speed is :

Space for your notes:

- (1) 4.25 km/h (2) 3.50 km/h
(3) 4.00 km/h (4) 3.75 km/h

Q9 - 31 January - Shift 2

A body is moving with constant speed, in a circle of radius 10 m. The body completes one revolution in 4 s. At the end of 3rd second, the displacement of body (in m) from its starting point is:

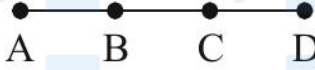
Space for your notes:

- (1) 30 (2) 15π
(3) 5π (4) $10\sqrt{2}$

Q10 - 01 February - Shift 1

#MathBoleTohMathonGo

An object moves with speed v_1 , v_2 , and v_3 along a line segment AB, BC and CD respectively as shown in figure. Where $AB = BC$ and $AD = 3 AB$, then average speed of the object will be :



(1) $\frac{(v_1 + v_2 + v_3)}{3}$

(2) $\frac{v_1 v_2 v_3}{3(v_1 v_2 + v_2 v_3 + v_3 v_1)}$

(3) $\frac{3v_1 v_2 v_3}{v_1 v_2 + v_2 v_3 + v_3 v_1}$

(4) $\frac{(v_1 + v_2 + v_3)}{3v_1 v_2 v_3}$

Space for your notes:

Q11 - 01 February - Shift 2

For a train engine moving with speed of 20 ms^{-1} . the driver must apply brakes at a distance of 500 m before the station for the train to come to rest at the station. If the brakes were applied at half of this distance, the train engine would cross the station with speed $\sqrt{x} \text{ ms}^{-1}$. The value of x is _____ (Assuming same retardation is produced by brakes)

Space for your notes:

(As per Official NTA Key released on 2 Feb)

Q4 (120)

Q8 (4)

Q11 (200)

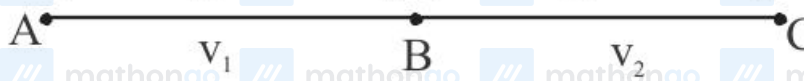
Q1 (4)

$$\text{Displacement} = \Sigma \text{area} = 16 - 8 + 16 - 8 = 16 \text{ m}$$

$$\text{Distance} = \Sigma |\text{area}| = 48 \text{ m}$$

$$\frac{\text{displacement}}{\text{Distance}} = \frac{1}{3}$$

Q2 (4)



$$\text{Average velocity} = \frac{\text{Total displacement}}{\text{Total time}}$$

$$= \frac{x + x}{\frac{x}{v_1} + \frac{x}{v_2}} = \frac{2v_1v_2}{v_1 + v_2}$$

Q3 (1)

$$x = 4t^2$$

$$v = \frac{dx}{dt} = 8t$$

$$\text{At } t = 5 \text{ sec}$$

$$v = 8 \times 5 = 40 \text{ m/s.}$$

Q4 (120)

$$\begin{aligned}
 v_i &= \sqrt{2gh_i} \\
 &= \sqrt{2 \times 10 \times 9.8} \downarrow \\
 &= 14 \text{ m/s} \downarrow \\
 v_f &= \sqrt{2gh_f} \\
 &= \sqrt{2 \times 10 \times 5} \uparrow \\
 &= 10 \text{ m/s} \uparrow
 \end{aligned}$$

$$|\vec{a}_{\text{avg}}| = \left| \frac{\Delta \vec{v}}{\Delta t} \right| = \frac{24}{0.2} = 120 \text{ m/s}^2$$

Q5 (1)

$$\frac{dx}{dt} = \text{slope} \geq 0 \text{ always increasing}$$

(A - II)

$$\frac{dx}{dt} < 0; \text{ and at } t \rightarrow \infty \frac{dx}{dt} \rightarrow 0$$

(B - IV)

$$\frac{dx}{dt} > 0 \text{ for first half } \frac{dx}{dt} < 0 \text{ for second half}$$

(C - III)

$$\frac{dx}{dt} = \text{constant}$$

Q6 (50)

A. mathongo B. mathongo C. mathongo mathongo mathongo mathongo mathongo

$$t_{AB} = \frac{x}{5 \text{ m/s}}$$

In motion BC

$$x = d_1 + d_2$$

where d_1 & d_2 we the distance travelled with 10 m/s and 15 m/s respectively in equal time intervals

't' each
 $\frac{t}{2}$

$$d_1 = \frac{10t}{2}, d_2 = \frac{15t}{2}$$

$$d_1 + d_2 = x = \frac{t}{2}(10 + 15) = \frac{25t}{2}$$

$$\langle v \rangle = \frac{2x}{\frac{x}{5} + \frac{2x}{25}} = \frac{2 \times 25}{5 + 2} = \frac{50}{7} \text{ m/s}$$

Ans. : 50

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Q7 (2)

Loss in PE = Gain in KE

$$\left(-\frac{GMm}{2R} \right) - \left(-\frac{GMm}{R} \right) = \frac{1}{2}mv^2$$

$$\Rightarrow v^2 = \frac{GM}{R} = gR$$

$$\Rightarrow v = \sqrt{gR}$$

Q8 (4)

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#MathBoleTohMathonGo

$$\frac{2}{V_{av}} = \frac{1}{3} + \frac{1}{5} = \frac{8}{15}$$

$$\Rightarrow V_{av} = \frac{15}{8} = 3.75 \text{ km/h}$$

Q9 (4)

No Solution

Q10 (3)

$$AB = x$$

$$BC = x$$

$$2x + CD = 3x$$

$$CD = x$$

$$\langle v \rangle = \frac{3x}{\frac{x}{u_1} + \frac{x}{u_2} + \frac{x}{u_3}} = \frac{3u_1 u_2 u_3}{u_2 u_3 + u_1 u_3 + u_1 u_2}$$

Q11 (200)

$$u = 20 \text{ m/s}, S_1 = 500 \text{ m}, v = 0$$

By third equation of motion

$$0 = (20)^2 - 2a \cdot 500 \Rightarrow a = \frac{4}{10} \text{ m/s}^2$$

$$u = 20 \text{ m/s}, S_2 = 250 \text{ m}, v = ?$$

$$v^2 = (20)^2 - 2a \cdot 250$$

$$= v = \sqrt{200} \text{ m/s}$$

$$x = 200$$